Week 02 Practice Problems (Optional)

2026-08-31

```
using Printf
```

Attempt these before Wednesday. These problems are optional; keep your work for yourself. Open the worked answer after you have tried the problem.

The module 1 written test draws from this pool.

Persistence from a linear reservoir

A linear reservoir has release constant $k = 0.3$. Its output under white-noise precipitation is an AR(1) process with coefficient $a = 1/(k + 1)$.

a. Compute a.
b. Compute the autocorrelation at lags 1, 2, and 5 using $\rho_h = a^{|h|}$.

Answer

$$a = \frac{1}{k+1} = \frac{1}{1.3}$$
$$\rho_1 = a$$
$$\rho_2 = a^2$$
$$\rho_5 = a^5$$

```
k = 0.3
a = 1 / (k + 1)
@printf("a = %.3g\n", a)
@printf("ρ₁ = %.3g, ρ₂ = %.3g, ρ₅ = %.3g", a, a^2, a^5)
```

```
a = 0.769
ρ₁ = 0.769, ρ₂ = 0.592, ρ₅ = 0.269
```

A single linear reservoir with one release constant turns white-noise precipitation into a persistent AR(1) output.

Estimation bias at short record length

The Kendall (1954) bias formula for the lag-1 autocorrelation estimated from a record of length n with unknown mean is

$$E(\hat{a}) \approx a - \frac{1 + 3a}{n - 1}$$

where a is the true value and $\hat{a}$ is the estimate (Mudelsee, 2010, pp. 56-57).

a. At $n = 50$ and $a = 0.7$, what is the expected estimate, and what percentage of the true value does the bias represent?
b. The bias-correction recipe inverts the formula: given an observed $\hat{a}$, solve for the corrected $\hat{a}'$. Show that $\hat{a}' = [\hat{a}(n-1) + 1]/(n-4)$, and verify that applying it to your answer from part a recovers $a = 0.7$.

ℹ Answer

Part a.

$$
\begin{aligned}
E(\hat{a}) &\approx 0.7 - \frac{1 + 3(0.7)}{50 - 1} \\
&= 0.7 - \frac{3.1}{49}
\end{aligned}
$$

```
a = 0.7
n = 50
bias = (1 + 3a) / (n - 1)
expected = a - bias
@printf("E(â) = %.3g, bias = %.3g, percentage = %.3g%%", expected, bias, 100 * bias / a)
```

```
E(â) = 0.637, bias = 0.0633, percentage = 9.04%
```

A 50-year record underestimates the persistence by about 9 percent.

Part b. In the bias formula, replace $E(\hat{a})$ with the observed $\hat{a}$ and a with the unknown $\hat{a}'$.

$$
\begin{aligned}
\hat{a} &= \hat{a}' - \frac{1 + 3\hat{a}'}{n - 1} \\
\hat{a}(n-1) &= \hat{a}'(n-1) - 1 - 3\hat{a}' \\
\hat{a}(n-1) + 1 &= \hat{a}'(n-4) \\
\hat{a}' &= \frac{\hat{a}(n-1) + 1}{n - 4}
\end{aligned}
$$

```
a_hat = expected
a_corrected = (a_hat * (n - 1) + 1) / (n - 4)
@printf("â = %.3g → â' = %.3g", a_hat, a_corrected)
```

```
â = 0.637 → â' = 0.7
```

The formula assumes a stationary AR(1) process with unknown mean and $|a| < 1$. It breaks down for a above about 0.9, where higher-order terms matter.

Variance of a time average under persistence

A 50-year climate record has lag-1 autocorrelation $a = 0.7$.

a. Compute the variance inflation factor $V = (1 + a)/(1 - a)$.
b. Compute the effective number of independent years $n' = n/V$.
c. A colleague tests whether the 50-year mean differs significantly from the historical baseline, treating the years as independent, and gets a test statistic of 2.0. By what factor is the test statistic inflated, and does the result survive the correction?

i Answer

Part a.

$$V = \frac{1+a}{1-a}$$

```
a = 0.7
V = (1 + a) / (1 - a)
@printf("V = %.3g", V)
```

```
V = 5.67
```

Part b.

$$n' = \frac{n}{V}$$

```
n = 50
n_eff = n / V
@printf("n' = %.3g effective independent years", n_eff)
```

```
n' = 8.82 effective independent years
```

Part c. The test uses σ_X^2/n for the variance of the mean, but the true variance is $V\sigma_X^2/n$. The standard error is $\sqrt{V}$ times larger than assumed, so the test statistic is $\sqrt{V}$ times too large.

```
t_stat = 2.0
inflation = sqrt(V)
corrected = t_stat / inflation
@printf("inflation factor = %.3g, corrected t = %.3g", inflation, corrected)
```

```
inflation factor = 2.38, corrected t = 0.84
```

The corrected test statistic is well below any conventional threshold. The apparent trend comes from persistence alone.

The large-sample formula $V = (1+a)/(1-a)$ assumes n is large enough that the finite-sample weights $(1-h/n)$ are close to 1. At $n = 50$ and $a = 0.7$ this is a reasonable approximation.

Reproducing a published autoregressive series

Example 9.6 of Wilks (2011) generates AR(1) and AR(2) series from one printed innovation sequence, starting from $x_0 = 0$. The first five innovations are $\varepsilon_1 = 1.526$, $\varepsilon_2 = 0.623$, $\varepsilon_3 = -0.272$, $\varepsilon_4 = 0.092$, $\varepsilon_5 = 0.823$.

The textbook writes ϕ_1 for the AR coefficient we have been calling a.

a. For the AR(1) model with $\phi_1 = 0.5$, compute x_1 through x_5 using $x_t = \phi_1 x_{t-1} + \varepsilon_t$.
b. The textbook also fits an AR(2) with $\phi_1 = 0.9$ and $\phi_2 = -0.6$, initializing $x_0 = x_{-1} = 0$. Compute x_1 for this model. The running text prints $x_1 = 1.562$. Does your arithmetic agree?

Answer

Part a.

$$x_1 = 0.5(0) + 1.526 = 1.526$$
$$x_2 = 0.5(1.526) + 0.623 = 1.386$$
$$x_3 = 0.5(1.386) - 0.272 = 0.421$$
$$x_4 = 0.5(0.421) + 0.092 = 0.303$$
$$x_5 = 0.5(0.303) + 0.823 = 0.974$$

```
ϕ = 0.5
ε = [1.526, 0.623, -0.272, 0.092, 0.823]
x = zeros(5)
x[1] = ε[1]
for t in 2:5
    x[t] = ϕ * x[t-1] + ε[t]
end
for t in 1:5
    @printf("x_%d = %.3g\n", t, x[t])
end
```

```
x_1 = 1.53
x_2 = 1.39
x_3 = 0.421
x_4 = 0.302
x_5 = 0.974
```

These match Table 9.4 of the textbook to three figures.

Part b. The AR(2) recursion is $x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \varepsilon_t$, with $x_0 = x_{-1} = 0$.

$$x_1 = 0.9(0) + (-0.6)(0) + 1.526 = 1.526$$

```
@printf("x₁ = 0.9(0) + (-0.6)(0) + 1.526 = %.3g", 0.9 * 0 + (-0.6) * 0 + 1.526)
```

```
x₁ = 0.9(0) + (-0.6)(0) + 1.526 = 1.53
```

The arithmetic gives 1.526, not 1.562. Table 9.4 also lists 1.526, and the next step in the textbook multiplies by 1.526, so the running text's 1.562 is a digit transposition. The rest of the worked example is correct.

Bibliography

Mudelsee, M. (2010). *Climate Time Series Analysis: Classical Statistical and Bootstrap Methods* (1 ed. 2010.). Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-90-481-9482-7

Wilks, D. S. (2011). *Statistical Methods in the Atmospheric Sciences* (3rd ed.). Amsterdam, The Netherlands ;: Elsevier.